INCH-POUND

MIL-W-44464
30 September 1992

MILITARY SPECIFICATION

WINTERGREEN BAR, FOOD PACKET, SURVIVAL, GENERAL PURPOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wintergreen bars for use by the Department of Defense as a component of the general purpose survival food packet.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATION

MILITARY

MIL-F-43231 - Food Packet, Survival, General Purpose, Packaging and Assembly of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

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NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeial (USP) and the National Formulary (NF)

(Application for copies should be addressed to the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Cellulose gel. The cellulose gel shall be composed of 15 parts of sodium carboxymethylcellulose and 85 parts microcrystalline cellulose. It shall be a white, water dispersible powder which shall retain not more than 1.0 percent on a U.S. Standard 60 mesh screen and not more than 30 percent on a U.S. Standard 200 mesh screen, and have an average particle size of 37 microns in dispersion. The microcrystalline cellulose shall have an ash content of not more than 5 percent and a heavy metals content of less than 10 parts per million (ppm). The pH of a 2 percent dispersion shall be not less than 6 nor more than 8, and shall have an acid demand requiring 2.2 mL of 0.1 normal hydrochloric acid to change the pH of 100 mL of a 1 percent solids dispersion to a pH of 4. The microcrystalline cellulose shall be insoluble in water and organic solvents, but shall be dispersible in water to form colloidal sols and gels (see 6.3.1).

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3.2.2 Flavoring, wintergreen, natural and artificial. The natural and artificial wintergreen flavor shall have a flavor typical of wintergreen and shall be in the oil form (see 6.3.2).

3.2.3 Magnesium stearate. Magnesium stearate shall be solid, (fine, bulky, white powder) with low bulk density, and shall possess a slight chracteristic flavor and odor. It shall be unctuous (grease-like) and free of grit. One hundred percent shall pass through a U.S. Standard 325 mesh sieve, and shall comply with the Food Chemicals Codex.

3.2.4 Sucrose, white, granular, direct compacting and tableting sugar. The tableting sugar shall be white in color and shall contain not less than 97 percent sucrose and may contain up to 3 percent maltodextrin. The tableting sugar shall have high flowability, low hygroscopicity, sweetness, shall be completely water soluble, and shall comply with the U.S. Pharmacopoeial and the National Formulary for compressible sugar. The sucrose shall be of such size that not more than 3.0 percent shall pass through a U.S. Standard 40 mesh screen and not less than 25 percent pass through a U.S. Standard No. 100 sieve (see 6.3.3).

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Wintergreen bars formulation and preparation. The wintergreen bars shall be formulated and prepared as follows:

Ingredients	Percent by Weight
Sucrose, white, granular, direct compacting and tableting sugar	98.30
Magnesium stearate	1.00
Cellulose gel	0.50
Flavoring, wintergreen, natural and artificial	0.20

NOTE: The following preparation procedures were used in the development of this product. Alternate procedures may be used provided finished product requirements are met. (When alternate procedures are used, the time requirements specified for the prepared product are still applicable.)

a. A small amount of sucrose shall be mixed with the natural and artificial wintergreen flavor until the flavor is well dispersed.

b. The remaining amount of sucrose and the cellulose gel shall be mixed in a suitable mixer for 4 minutes.

c. The sucrose/wintergreen flavor mixture shall be added to the sucrose/cellulose gel mixture and mixed for 2 minutes.

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d. The magnesium stearate shall be added to this mixture and mixed for 1 minute.

3.3.2 Compression of wintergreen bars. The powder mixture shall be compressed so as to produce bars complying with the finished product requirements.

3.4 Pouch filling and sealing. The compressed bars shall be packaged immediately in accordance with MIL-F-43231.

3.5 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odors or flavors such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. There shall be no compression streaks or stains.

e. Each bar shall be 1.0 ± 0.2 inches wide by 3.0 ± 0.1 inches long by 1.0 ± 0.1 inches thick.

f. The weight of each bar shall be not less than 35 grams and not greater than 45 grams.

g. The moisture content shall be not less than 1.0 percent and not greater than 3.0 percent.

h. The color of the wintergreen bar shall be white.

i. The product shall have an odor and flavor typical of wintergreen.

j. The compressed bar shall not be fragile or excessively hard.

k. No pouch shall contain a bar which has been broken or crumbled.

l. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample.

m. The E. coli count shall be less than 3 per gram (no positive tubes in the standard three tube MPN technique) in four of five samples and not greater than 20 per gram in any sample.

3.5.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

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3.6 Plant qualifications. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110: "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food," and the plant sanitation requirements of the appropriate Government inspection agency.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each

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ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the AOAC, and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, and formulation, or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch examination. Examination of the filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-F-43231.

4.5.4 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 35 grams or greater than 45 grams shall be classified as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 1 gram.

4.5.5 Product examination. The product shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE I. Product defects 1/ 2/

Category		Defect
Major	Minor	
101		Product does not possess the odor and flavor typical of wintergreen
102		Presence of compression streaks or stains
103		Dimensions of bar not as specified
104		Bar is fragile or excessively hard

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TABLE I. Product defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
	201	Color of bar not as specified
	202	Broken or crumbled bar 3/

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.5.1).

3/ A broken or crumbled bar is a bar which either:

a. Has been broken into more than three pieces, the smallest of which is larger than 1/2 cubic inch in volume; or

b. Has more than 1/8 of the volume of the bar crushed.

4.5.6 Moisture content testing. Eight filled and sealed pouches shall be randomly selected from the lot regardless of lot size. The finished product shall be individually tested for moisture content in accordance with the Official Methods of Analysis of the AOAC, method 925.45A. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the moisture content requirements in 3.5 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Microbiological testing. Five filled and sealed pouches shall be selected at random from the lot regardless of lot size. The finished product shall be individually tested for microbiological levels in accordance with the AOAC method 966.23. The diluent shall be added to each bar and allowed to stand for 15 minutes before blending the sample. Any result failing to conform to the microbiological requirements in 3.5 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.8 Packaging examination. Examination of packaging shall be in accordance with the quality assurance provisions of MIL-F-43231.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-F-43231.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.5.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Ingredient information.

6.3.1 Cellulose gel. It has been found that Avicel Microcrystalline Cellulose, NF, Type PH-101, manufactured by FMC Corporation - Food & Pharmaceutical Products Division, 1735 Market St., Philadelphia, PA 19103 meets the requirements of 3.2.1 and performs satisfactorily in this product.

6.3.2 Flavoring, wintergreen, natural and artificial. It has been found that Natural and Artificial Wintergreen No. 322610T manufactured by F&C International, 890 Redna Terrace, Cincinnati, OH 45215 meets the requirements of 3.2.2 and performs satisfactorily in this product.

6.3.3 Sucrose, white, granular, direct compacting and tableting sugar. It has been found that Di-Pac manufactured by Amstar Corporation, American Sugar Division, 1251 Avenue of the Americas, New York, NY 10020 meets the requirements of 3.2.4 and performs satisfactorily in this product.

6.4 Subject term (key word) listing.

Combat field feeding
Compressed food
Food processing
Tableting sucrose

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Custodian:

Air Force - 50

Review activities:

Army - MD, QM
Navy - MC, SA
DP - SS

Preparing activity:

Army - GL

(Project 8940-0733)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-W-44464	1992 September 30

3. DOCUMENT TITLE

WINTERGREEN BAR, FOOD PACKET, SURVIVAL, GENERAL PURPOSE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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